Can LLMs Uncover Hidden Physical Laws through Free Exploration?

Anonymous ACL submission

Abstract

Scientific discovery often begins with an unexpected observation that prompts a new research question. In a world with altered physical laws, can large language models (LLMs) notice unexpected behavior, such as heavier objects falling more slowly, and discover the law through experiments without being told what to investigate? To evaluate this ability, we introduce LongChang, an automated framework that creates simulated worlds with altered physics, where one or more laws differ from ordinary physics. The evaluated LLMs act as explorers, manipulating objects and building tools to investigate their own questions without research tasks or hints about these laws. Separate LLM agents generate and review scenes, review tool designs, and evaluate reports for correctness and experimental support using the altered physical laws and exploration records. These records guide scene evolution for further exploration, allowing the full cycle to run without human intervention. Among 12 LLMs, GPT-6-Astra achieves a discovery rate of 16.7%, while the other 11 score 0%. The exploration records reveal uninvestigated anomalies and only partial identification of the altered physical laws. Our anonymized materials are available here.

1 Introduction

Large language models (LLMs) aid scientific discovery through hypothesis generation, experiment design, and evidence analysis. Scientific discovery often begins with an unexpected observation that prompts a new research question. To pursue such inquiry autonomously, LLMs must decide what to investigate and test their explanations through experiments. This ability could help scientists investigate unexpected phenomena and pursue questions they might otherwise overlook. LongChang's name reflects independent inquiry, inspired by the philosopher Wang Yangming's introspection in a cave at Longchang, where he reconsidered how to seek knowledge and reached an insight that helped shape his own School of Mind (Yangming, 2026).

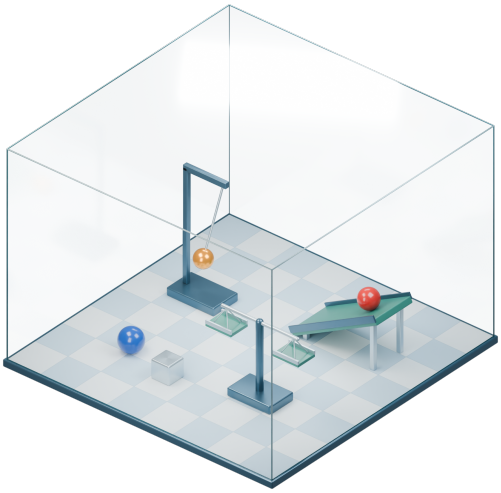


Figure 1: LongChang's glass room. Explorers manipulate objects and build tools. Other LLMs evolve scenes using exploration records without human intervention.

Physical law discovery involves inferring an environment's laws from observations and experiments. PhysGym (Chen et al., 2025), NewtonBench (Zheng et al., 2026), and DiscoverPhysics (Wiemann et al., 2026) assign law-discovery tasks. Tab. 2 compares their exploration guidance and experiments with those of LongChang. Without a research task or hints about altered physical laws, an LLM must notice phenomena worth investigating, choose a research question, and devise experiments to distinguish competing explanations. Can LLMs turn unexpected observations into research questions, conduct experiments, and discover altered physical laws autonomously?

We introduce LongChang, an automated framework that generates simulated worlds with altered physics, where one or more laws differ from or-

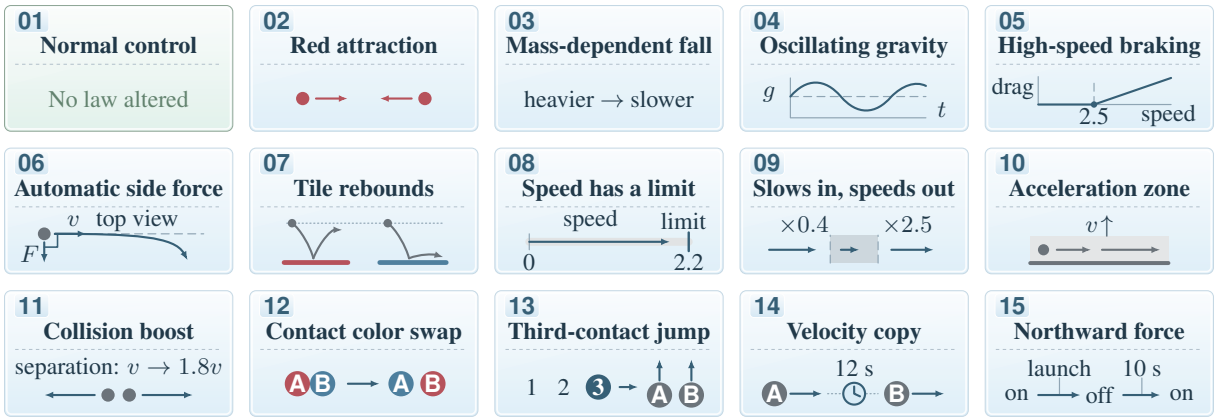


Table 1: Scene 01 uses ordinary physics. Scenes 02–15 each have one altered physical law. Laws are in App. A.1.

dinary physics. This counterfactual setting tests whether LLMs independently notice departures from ordinary physics and investigate them without research tasks or hints about the altered physical laws. For example, slower falls for heavier objects could lead an explorer to ask how mass affects falling and design controlled comparisons to test this link. These laws are hidden from explorers but available for evaluating their reported explanations.

LongChang's base set in Tab. 1 comprises a normal control and 14 scenes with one altered physical law each. In the virtual "glass room" in Fig. 1, explorers manipulate objects and build tools that interact with the scene objects within a shared MuJoCo rigid-body simulation (Todorov et al., 2012). Separate LLM agents review tool designs and evaluate reports against the altered physical laws and exploration records, requiring both correctness and experimental support for discovery. Unresolved questions and corrected errors in these records guide LLM scene designers to generate, refine, and compose scenes for further exploration, continuing the entire discovery cycle without human intervention.

Among 12 evaluated LLMs, GPT-6-Astra achieves a discovery rate of 16.7% under free exploration, while the other 11 score 0%. Exploration records show some LLMs leave observed anomalies uninvestigated, while others experiment but only partially identify the altered physical laws, leaving their conditions or applicability unresolved.

We contribute the LongChang framework for the complete discovery cycle and an empirical study of autonomous physical inquiry. We use model comparisons, controlled interventions, and exploration records to examine how LLMs form research questions and test explanations using experiments.

Closest work	No task	Interactive 3D scenes	Tool design	Scene evolution
PhysGym	✗	✗	✗	✗
NewtonBench	✗	✗	✗	✗
DiscoverPhysics	✗	✗	✗	✗
LongChang	✓	✓	✓	✓

Table 2: LongChang compared with the closest work: PhysGym (Chen et al., 2025), NewtonBench (Zheng et al., 2026), and DiscoverPhysics (Wiemann et al., 2026). No task: no assigned research goal. Interactive 3D scenes: physical worlds in MuJoCo (Todorov et al., 2012). Tool design: devices built to interact with scene objects. Scene evolution: automatically generating and revising scenes guided by exploration records.

2 Related Work

2.1 Autonomous Scientific Discovery

PhysGym (Chen et al., 2025) and NewtonBench (Zheng et al., 2026) ask LLMs to recover hidden equations through numerical queries, while DiscoverPhysics (Wiemann et al., 2026) studies law discovery in particle simulations. These benchmarks specify discovery goals before experimentation, as summarized in Tab. 2. Other protocols require explicit initial hypotheses and a minimum number of experiments (Nägele and Marquardt, 2025) or supply the anomaly to explain (Geng et al., 2026). Open-ended systems let LLMs select questions within an assigned research role, using supplied datasets (Agarwal et al., 2025; Chen et al., 2026; Yao et al., 2025) or an optical research setting (Zeng et al., 2026; Yang et al., 2026). LongChang tests whether LLMs notice unusual effects of altered physical laws, initiate investigations, and establish these laws experimentally, without a preassigned research task or hints about these laws.

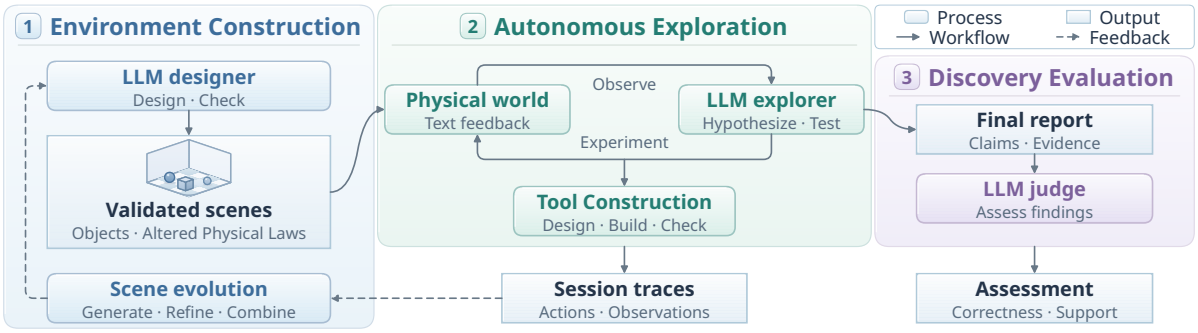


Figure 2: LongChang's automated cycle of scene construction, free exploration, assessment, and scene evolution.

Left Alone (Szeider, 2025) studies self-directed behavior without an assigned task or a simulated physical world. Explore-then-Act in Look Before You Leap (Ye et al., 2026) studies task-free exploration, including in ScienceWorld, with modeling instructions and a warning that expectations may fail. It evaluates exploration and downstream task performance, rather than experimentally supported discovery of altered physical laws by the explorer.

Both WorldTest (Warrier et al., 2025) and OneLife (Khan et al., 2026) examine predictive world models and explicitly ask explorers to learn environment rules. LongChang instead asks whether unexpected physical observations prompt LLMs to pose their own questions and establish altered physical laws through experiments, without research tasks or advance hints. Tab. 2 therefore focuses on frameworks for physical law discovery.

2.2 Tool Use and Design

DiscoveryWorld supplies instruments and abstracted object interactions in a grid world (Jansen et al., 2024), while ChemWorld exposes chemical procedures through programmable modules (Qiu et al., 2026). Tool-design systems optimize tool geometry and use for a specified manipulation task (Lin et al., 2025) or propose detector layouts under given performance goals and resource constraints (Zoccheddu et al., 2026). LongChang ties tool construction to the explorer's own investigation: LLMs decide whether a tool is needed, what to build, and how to use it to test a hypothesis. The explorer's constructed tools interact physically with the other objects in the same simulated world.

2.3 Environment Generation and Evolution

Environment-generation methods create tasks and rewards for agent training and adapt environments based on performance or trajectory feedback (Faldor et al., 2025; Kang et al., 2026). Programmable chemical worlds support controlled changes to private laws while preserving public experimental conditions (Qiu et al., 2026). Trajectory-based benchmark construction uses tool-use records to create new tasks with solutions checked by re-execution (Guo et al., 2025). In LongChang, LLM scene designers use unresolved questions and corrected errors in exploration records to generate, refine, and compose scenes. Explorer LLMs choose observations to investigate, while separate LLM agents review tools and assess reports, continuing this discovery process without human intervention.

3 LongChang

Fig. 2 presents LongChang's automated cycle of physical law discovery: environment construction (Sec. 3.1), autonomous exploration (Sec. 3.2), and discovery evaluation (Sec. 3.3). The evaluated LLMs serve as explorers, while separate LLM agents design and review scenes, review apparatus, and evaluate reports. LLMs carry out the entire loop without human intervention, using exploration records to guide the subsequent evolution of scenes.

3.1 Environment Construction

Physical Environment. An LLM designer specifies each scene's objects and geometry, physical program, and any altered physical laws. The framework compiles these definitions into MuJoCo simulations (Todorov et al., 2012) containing manipulable balls and boxes, with colored floor tiles in some scenes. Each base scene with altered physics modifies one physical law, producing effects that depend on object properties, spatial relations, contact events, or past actions. Fig. 3 illustrates an altered physical law: when the balls are close, both rise under an upward force without an upward launch.

The designer records each altered physical law's

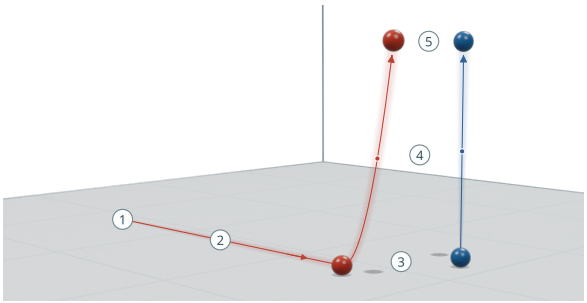
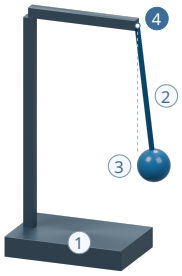


Figure 3: An example of an altered physical law. When the two balls are close, an upward force makes them rise together. Labels 1–5 identify successive snapshots.



	Component	Type
1	Support	3 boxes
2	Rod	capsule
3	Bob	sphere
4	Hinge	joint
	Hinge angle	–8°

Figure 4: An example of an LLM-built tool. The pendulum uses its hinge angle to test whether local gravity is tilted. The instrument reads $-8°$ in the illustrated state.

target objects, conditions, and observable effects for evaluation, withholding this information from the explorer. LongChang's base set in Tab. 1 comprises a normal control and 14 scenes with one altered physical law each. The normal control assesses whether LLMs ground their conclusions in evidence when no physical law has been altered. App. A.1 details ordinary physics and each altered physical law's scope, conditions, and parameters.

Scene Evolution. To extend this set, an LLM designer proposes scene changes aimed at more challenging discovery tasks. The designer receives source scenes and any available summaries of earlier explorations. These summaries identify tested conditions, unresolved questions, and explanations supported or corrected by later observations. The designer uses three operations: generation creates new laws and scenes from source mechanisms, refinement modifies an existing scene, and composition combines source laws in a shared environment. Changes guided by these summaries must cite supporting exploration evidence. Automated tests check simulated behavior against the altered physical laws, including interactions with constructed apparatus. An LLM reviewer checks the program and test results against these laws. Failed compilation, tests, or review trigger automatic revision requests to the designer. Only accepted scenes are used in subsequent free-exploration assessments.

3.2 Autonomous Exploration

The explorer chooses which observations to investigate, formulates its own research questions, and tests explanations through physical interventions. The initial prompt asks for free exploration without a research task or hints about the altered physics. Structured observations give object positions, motion, and instrument readings, with distances in bricks and elapsed time in counts. The explorer can push or throw objects, grasp and move them, release them, wait, and mark reference locations. App. A.2 documents the complete action interface.

When an investigation needs additional control or measurement, the explorer can design apparatus such as the pendulum in Fig. 4. It specifies the device's geometry and mechanical connections, including joints and springs. An LLM reviewer checks the compiled design for physical validity and returns feedback on rejections. The explorer can revise a rejected design and submit it again, with successful designs installed into the current scene. Once installed, the explorer can place objects on the apparatus or adjust its joints and receive updated motion readings with the next observation.

At each turn, the explorer records its observations and action rationale, then acts and receives updated observations. The cycle continues until it stops or reaches the turn limit, retaining notes, actions, observations, and feedback from all attempts.

The final report summarizes the observations and explanations from this record. Each conclusion must cite supporting turns and use only interpretations already recorded in the notes, preserving the uncertainty stated there. The explorer may also report that it found no anomaly during exploration.

3.3 Discovery Evaluation

An LLM judge evaluates reports against altered physical laws and full exploration records, requiring law correctness (C) and evidence support (S) for discovery. Law correctness requires identifying each altered physical law, its target objects, conditions, triggers, and constants identifiable in the reported units. Proposed mechanisms may differ if they yield the same observable predictions under allowed interventions. Evidence support requires recorded experiments to justify all claims and reasoning at the stated level of certainty. Cor-

	Discovery (%)			Control	Exploration				
Model	D	C	S	R	L	ES (%)	CS	CF (%)	AF (%)
GPT-5.6-Luna	0.0	0.0	38.1	0	36.7	95.6	0	–	0.8
GPT-6-Astra	16.7	19.0	97.6	0	39.3	100.0	11	48.0	2.1
Claude-Opus-5	0.0	16.7	0.0	0	85.8	100.0	44	66.6	18.2
Gemini-3.7-Flash	0.0	0.0	19.0	0	10.9	100.0	1	75.0	6.1
GLM-5.3	0.0	4.8	0.0	3	91.6	97.8	45	64.0	16.8
GLM-5.3-Flash	0.0	4.8	0.0	1	87.6	95.6	45	71.4	20.2
Kimi-K3	0.0	2.4	0.0	3	83.9	93.3	42	82.1	32.7
MiMo-V2.6-Flash	0.0	4.8	0.0	1	36.2	100.0	33	71.1	24.2
MiMo-V2.6-Pro	0.0	7.1	0.0	2	42.3	100.0	35	70.1	19.5
DeepSeek-Flash	0.0	0.0	0.0	2	45.6	100.0	42	77.1	19.2
Qwen3-Max	0.0	0.0	9.5	0	26.7	100.0	16	63.8	6.4
Grok-4.7	0.0	0.0	2.4	1	85.2	24.4	2	88.5	12.2

Table 3: Discovery performance and exploration behavior of 12 LLMs in LongChang, with metrics from Sec. 3.4.

rect guesses may pass only C. Supported ordinary observations or partial findings may meet S alone.

3.4 Experimental Setup

Main evaluation. All LLMs explore the same 15 scenes from fixed initial states for up to 100 turns.
LLMs. We evaluate the 12 LLMs in Tab. 3. GPT-6.1-Sol judges the correctness and experimental support of their reports. GPT-6-Astra reviews apparatus and generates and reviews new scenes. App. A.3 lists the prompts used across the pipeline.
Metrics. For scenes with altered physics, C and S show the percentages of runs passing the relevant check. Discovery rate (D) is the percentage passing both checks. Rejection count (R) is the number of rejected normal-control reports, including those with false claims of anomalies or unsupported laws.
Exploration statistics. Across all scenes, exploration length (L) is the mean number of turns used, and early stopping (ES) is the percentage of runs ending before the turn limit. Action failure (AF) is the percentage of turns with failed actions. Construction success (CS) counts explorations with at least one successful build, and construction failure (CF) gives the percentage of failed build attempts. These statistics distinguish budget use, access to working apparatus, and action execution reliability.

4 Results and Analysis

Tab. 3 shows that GPT-6-Astra achieves a discovery rate of 16.7% under free exploration, while the other 11 LLMs score 0%. Its evidence support rate reaches 97.6%, but law correctness is only 19.0%: supported observations and partial findings often leave the altered physical laws unresolved. For example, a Scene 02 report supports attraction between the tested red balls without establishing its distance dependence or applicability to other red spheres. Claude-Opus-5 shows the opposite pattern, identifying the altered physical laws in 16.7% of runs while no report passes the evidence support check. These results separate correct identification of a law from experimentally supported discovery.

Exploration statistics reveal obstacles during investigation. Gemini-3.7-Flash stops early in every run, averaging only 10.9 turns and leaving little opportunity for follow-up experiments. The GLM models use about 90 turns and build working apparatus in every run, yet make no discoveries. They also fail in more than 60% of build attempts, with failed actions accounting for 16.8–20.2% of turns. Successful construction can thus coexist with repeated unsuccessful attempts. Grok-4.7 stops early least often but makes no discoveries, so persistence and working devices alone do not ensure discovery.

4.1 Forming Research Questions

We study whether explorers turn unexpected observations into research questions and pursue them experimentally. Fig. 5 compares two Scene 15 explorations with the same initial state and prompt. When the red ball starts moving while the blue ball is held, GPT-6-Astra postpones its planned collision test to investigate. It measures the restart delay and tests whether pushing blue also resets it, discovering that pushes or throws pause red's northward force for ten counts. Gemini-3.7-Flash observes a similar restart but continues its planned tests, leaving the unexpected motion unexamined.

We test guidance by giving GPT-6-Astra a law-discovery task, noting that physical laws may be altered, and supplying a scene-specific clue, with other settings fixed. In Tab. 4, discovery rises from 16.7% to 28.6%, while evidence support remains

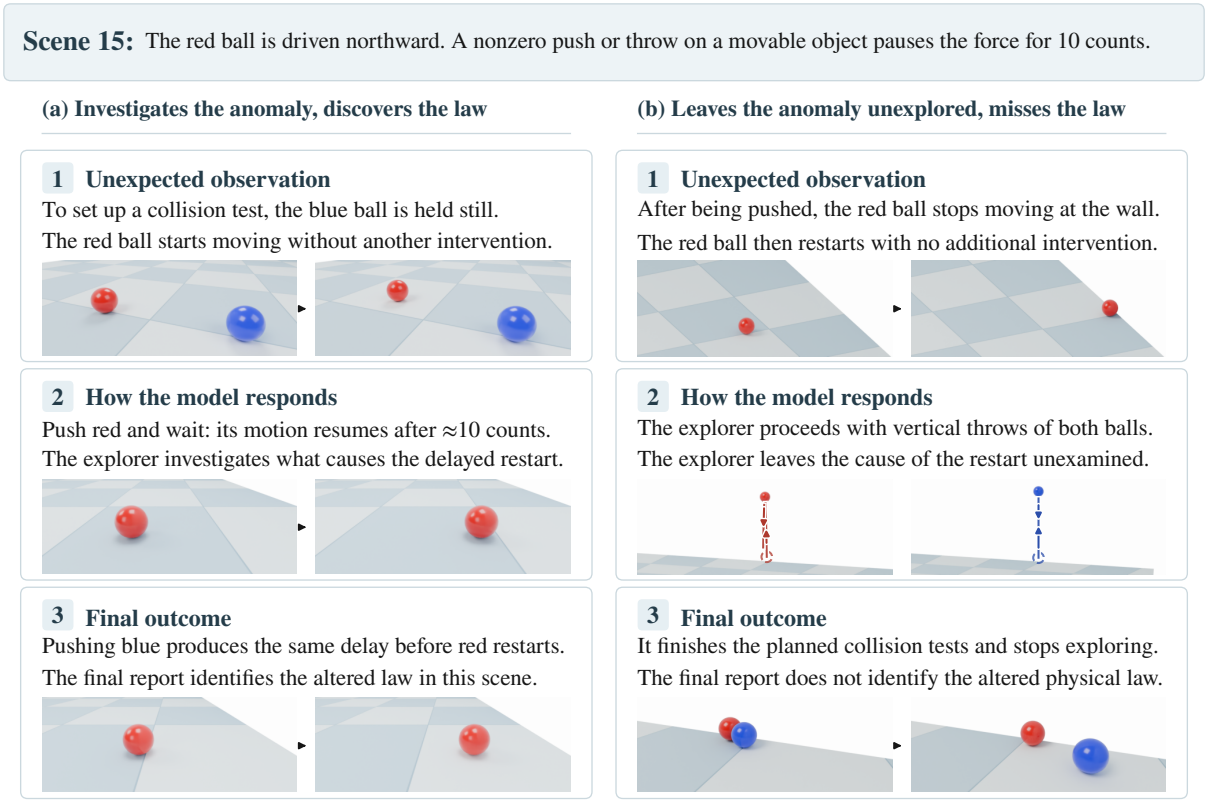


Figure 5: Case study of research question formation in Scene 15. (a) GPT-6-Astra investigates red's delayed restart and discovers the altered physical law. (b) Gemini-3.7-Flash observes the restart but still follows its original plan.

Condition	D (%)	C (%)	S (%)	L	AF (%)
Task guidance, 100 turns					
Free exploration	16.7	19.0	97.6	39.3	2.1
Task-guided	28.6	31.0	97.6	30.9	2.7
Turn budget					
100 turns	16.7	16.7	100.0	42.2	2.7
500 turns	14.3	14.3	100.0	41.4	3.6
Apparatus access, 100 turns					
Enabled	16.7	19.0	97.6	39.3	2.1
Disabled	4.8	4.8	100.0	38.7	0.9

Table 4: Effects of task guidance, turn budget, and apparatus access on discovery and exploration behavior.

unchanged. Guidance improves law identification when experimental support is already high. Mean exploration length falls from 39.3 to 30.9 turns, yielding more discoveries with fewer interactions.

4.2 Validating Scientific Claims

Distinguishing competing explanations requires experiments for which they predict different outcomes. Fig. 6 illustrates this process in Scene 12. Two successive red–blue collisions change and restore the balls' colors, consistent with both color exchange and independent red–blue flipping. GPT-6-Astra tests two red balls, for which exchange predicts that both remain red and flipping predicts that both turn blue. Both balls remain red, supporting exchange and ruling out independent flipping.

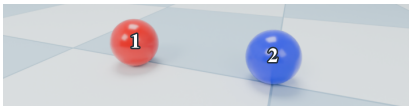
The normal control tests whether LLMs overinterpret observations under ordinary physics. Across LLMs, 36.1% of reports are rejected for false claims of altered physical laws or unsupported explanations of ordinary physics. For example, DeepSeek-Flash attributes a ball's longer travel on the decorative gray tile to lower friction after comparing equal pushes. The ball is already moving before each push, so different starting speeds could explain the difference in distance. A fair comparison would start the ball from rest before each push.

4.3 Using Experimental Resources

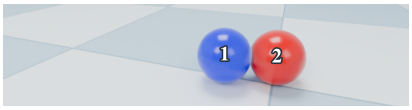
We examine whether larger budgets or apparatus access helps LLMs resolve unanswered questions. We compare budgets of 100 and 500 turns under free exploration without context compaction, with other settings matched. Increasing the limit to 500 turns does not improve discovery: GPT-6-Astra uses about 40 turns under either limit and stops early, sometimes leaving questions unresolved. Ex-

Scene 12: Two balls swap colors once when they touch. They must move apart and touch again for another swap.

1 Exploratory test
Before contact: a pair of balls, one red and one blue



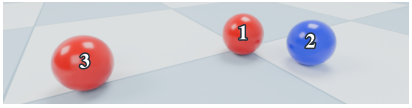
2 Unexpected observation
Contact changes the colors of both balls in the test. Repeated contact restores both balls' original colors.



3 Competing explanations
Both explanations fit the observed changes in color. A new experiment is required to distinguish the two.

Exchange colors	1	2
Flip each color	1	2

4 Design a discriminating test
Test two red balls to distinguish these explanations.



5 Conflicting predictions
The two explanations predict different test outcomes.

Exchange colors	3	1
Flip each color	3	1

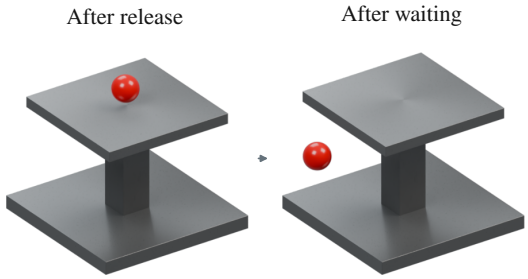
6 Test the competing explanations
Both balls stay red, which supports color exchange. Flipping predicts that both turn blue and is ruled out.



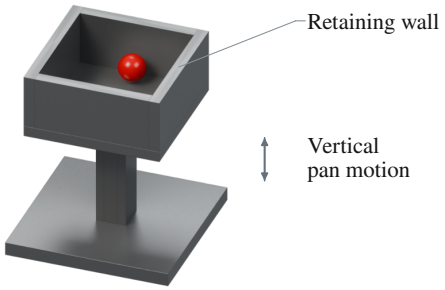
Figure 6: Case study of testing competing explanations.

Scene 15: The red ball experiences a northward force. Any nonzero push or throw pauses this force for 10 counts.

1 Flat pan: measurement fails
Without retaining walls, the released red ball drifts off. The drift prevents a stable load reading for the red ball.



2 Redesign and measure
Add retaining walls and extend the pan's vertical travel.



3 Measurements in time order
Pan drop from first empty reading (bricks)

1	Empty pan	0.0
2	Add and release red	0.1
3	Remove red (empty)	0.0
4	Add and release blue	0.1

Figure 7: Case study of tool construction and redesign.

tra turns provide little value if explorers stop early.

The apparatus comparison uses a 100-turn limit. Disabling construction removes its actions and instructions but retains basic object manipulation. With construction enabled, discovery rises from 4.8% to 16.7% and law correctness from 4.8% to 19.0% in Tab. 4, although action failures also increase. The explorer chooses whether to build a tool, so these results concern access to construction rather than the contribution of tool use in each run.

Fig. 7 illustrates the construction and redesign of a weighing platform in Scene 15. GPT-6-Astra builds a flat pan to compare the balls' loads, but the red ball drifts off after release. It adds retaining walls and increases the pan's vertical travel, obtaining stable readings after releasing each ball.

5 Scene Evolution

Exploration records can also inform what LLMs investigate in subsequent scenes. We study whether corrected errors from successful explorations can guide generation, refinement, and composition of new discovery challenges. Each operation modifies a mechanism involved in the same error to create a scene with altered physics for further exploration.

We review successful GPT-6-Astra explorations for explanations corrected after later observations. In Scene 15, red stops after a push, prompting the

Scene 15: A nonzero push or throw pauses the red ball's northward force for 10 s.

Corrected error. The explorer initially interpreted this pause as evidence of no lasting force.

How the scene evolves	Altered physical laws
Generation	
Replace the northward force with mutual repulsion between the red and blue balls, retaining the 10 s pause.	The red and blue balls are subject to mutual horizontal repulsion. Any nonzero push or throw pauses this mutual repulsion for 10 s.
Refinement	
Block the red ball's northward force while the blue ball is inside the gray square, even after the 10 s pause ends.	Blue inside the gray square blocks the red ball's northward force. Nonzero pushes or throws also pause the northward force for 10 s.
Composition	
Add Scene 14's rule: the green ball copies red's launch velocity 12 s after a nonzero push or throw acts on red.	A nonzero push or throw on any object pauses red's force for 10 s. Each red launch makes green copy red's launch velocity 12 s later.

Table 5: Scene evolution guided by a corrected error in Scene 15, with altered physical laws in three new scenes.

Scene	C (%)	S (%)	D (%)
Scene 15	33.3	100.0	33.3
Scene 14	66.7	100.0	66.7
Generation	0.0	100.0	0.0
Refinement	0.0	100.0	0.0
Composition	0.0	100.0	0.0

Table 6: Discovery rates in source and evolved scenes.

inference that no persistent northward force exists. When red restarts, the explorer revises this explanation to account for a temporary pause in the force. The designer receives this error, its correction, and the source laws, and uses the same evidence for all three operations. Tab. 5 shows how the three designs retain the pause while modifying these laws.

Generation. The designer replaces the fixed northward force on red with mutual repulsion between red and blue, making the force depend on their relative positions. Swapping their positions at matched times after a launch should reverse the forces, distinguishing repulsion from a fixed directional force.

Refinement. The designer introduces a spatial condition blocking red's northward force while blue is inside the gray square, so waiting longer than ten seconds may still not restore it. After the pause expires, the planned test moves blue by hand into or out of the square without resetting the timer.

Composition. The designer combines the paused northward force with Scene 14's delayed motion copying: green takes red's launch velocity twelve seconds later. The planned comparison launches blue after red, postponing red's force recovery while leaving green's scheduled velocity copy unchanged. The explorer must identify both altered physical laws and their distinct triggers and timing.

We evaluate GPT-6-Astra on source and evolved scenes using the free-exploration settings in Tab. 3.

Tab. 6 reports discovery rates of 33.3% in Scene 15 and 66.7% in Scene 14, compared with 0% in all three evolved scenes. Evidence support remains at 100%, but no evolved-scene report passes law correctness. In the generated scene, an exploration identifies position-dependent repulsion without recovering its pause and delay. In the refined scene, the explorer observes northward motion without linking it to blue's position. In the composed scene, it records red's drift and green's intermittent motion without identifying the two timing rules. These cases use a corrected error to guide scene evolution, revealing unresolved conditions and dependencies in otherwise supported findings.

6 Conclusion

LongChang examines whether LLMs can turn unexpected observations into research questions and discover altered physical laws through experiments. The evaluated LLMs manipulate objects and build tools without research tasks or hints about these laws. Only one of the twelve evaluated LLMs makes discoveries, revealing the difficulty of turning observations into complete, experimentally supported laws. Unanswered questions and corrected errors in exploration records guide separate LLM agents to evolve scenes, continuing the discovery cycle without human intervention. This framework supports research into LLMs that decide what to investigate and design experiments to resolve uncertainties about the physical laws of these worlds.

7 Limitations

This study examines whether LLMs initiate investigations and discover altered physical laws in simulated worlds. Our results show that this task remains challenging for current LLMs. The altered physical laws provide a clear basis for evaluating discoveries, while many scientific questions in the real world remain unresolved. Extending the framework to these open questions requires further research on evaluating exploration and its findings in settings where the answers are not yet established.

8 Ethics Statement

LongChang uses no human annotation or testing and evaluates LLMs without human intervention. The evaluation therefore raises no ethical concerns related to participant privacy or informed consent.

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A Appendix

A.1 Scene Laws

Scene 01 provides the normal control: three movable balls and a decorative gray tile in a room with ordinary rigid-body physics. Motion follows $\mathbf{F} = m\mathbf{a}$ and $\mathbf{p} = m\mathbf{v}$, with constant downward gravitational acceleration independent of mass. Contact with other objects, the floor, and walls produces collisions and friction. Object colors remain fixed, and the gray tile has no physical effect on motion.

Scenes 02–15 each change one physical law of this setting. Tab. 7 and Tab. 8 detail the laws and scope of the 15 base and the three evolved scenes.

A.2 Action Interface

Tab. 9 specifies the target objects, parameters, and effects of the actions available during exploration.

A.3 Prompts

The prompts below guide exploration, generate final reports, and evaluate discoveries. Task guidance supplements the original exploration prompt.

A.4 AI Assistance

We used AI assistants to improve the wording and clarity of the manuscript and to assist with code implementation and debugging. We also used GPT-6.1-Sol as an automated judge to assess the correctness and experimental support of exploration reports, as described in Sec. 3.3. The authors are responsible for the manuscript, code, and findings.

A.5 Potential Risks

LLM judges may endorse unsupported findings and mislead subsequent research. Our experiments are simulated, but physical use of generated experiments or apparatus could cause harm without independent validation and expert safety review. Repeated LLM inference and simulation also consume energy, with associated environmental costs.

Free exploration

```
Explore freely and report what you learn.

## Basic operations
- PUSH <object> <heading> <impulse>.
- THROW <object> <heading> <elevation> <impulse>.
- WAIT <counts>: advance the world by 1-30 whole
counts.
- GRASP <object>: hold.
- MOVE <object> <x> <y> <z>: move held object.
- RELEASE <object>: let go.
- MARK <id> <x> <y>: place a floor reference.

## Parameters and observations
- Heading: degrees (0=east, 90=north) or
east/north/west/south; elevation: 0-90 degrees.
- Impulse: 0-10, added to current motion.
Positions: bricks.
- RELEASE a held target before PUSH/THROW.
- Object motion is in events[name].trajectory:
count, position, state. Counts are relative to the
action start and may be fractional; intervals may
differ. No absolute time calibration is supplied.
- omitted_samples counts compressed samples;
key_points_omitted warns that key changes were
dropped.
```

```
- Use provided objects/observations, obey
usage/handling limits, never invent results.

## Reply
NOTE: <observation and reason>
ACT: <one command, DONE, or empty>
- One NOTE line, one ACT field; no Markdown;
multiline only for BUILD JSON.
- DONE ends exploration early and requests a
report; empty ACT skips physical action.

## Optional construction
- You may create apparatus with BUILD; state its
intended use in NOTE.
- BUILD <JSON>: submit apparatus; required id,
purpose, mjcf; optional name.
- PLACE <object> <tool> <port>: carry and hold an
object at a port.
- ADJUST <tool> <joint> <target>: use a listed
hinge/slide joint; target units are degrees for
hinges, bricks for slides.
- Design your own bodies, geometry, mass/density,
friction, joints and sites; no preset part
catalogue.
- mjcf: complete XML
<mujoco><worldbody>...</worldbody></mujoco>.
Sections: worldbody with bodies, optional compiler
(angle only) and tendon.
- Allowed geoms: sphere, box, capsule, cylinder,
ellipsoid; joints: free, ball, hinge, slide.
Joint/tendon springs allowed; tendons need finite
limits and may refer only to this apparatus.
- Unsupported: assets, includes, defaults,
actuators, sensors, mocap, gravcomp.
- Root positions: world coordinates; nested:
relative. Box sizes: half-extents. Angles: degrees
by default, or <compiler angle="radian"/>.
- Names: lowercase letters, digits, underscores.
Local names gain tool_<id>_ prefix. Use returned
object/joint IDs.
- Each site provides a port and position readout;
hinges provide angles, slides travel. Readings are
returned automatically in each observation's
instruments[].state.
- No quota on apparatus count, component count or
material mass. BUILD requires compilation,
independent physical review and installation checks.
- Successful BUILD installs the apparatus; its
readouts appear in subsequent observations.
```

Final report

```
# Exploration record
Summarize received observations, including the
final observation. Include only interpretations
already expressed in your exploration notes,
preserving their uncertainty. Do not introduce new
explanations or discoveries, or retrospectively
claim that an action was intended to test a
question you had not raised during exploration.

## Output format
Return one JSON object with exactly this structure
and no surrounding text:
{
  "experiments": [ {
      "turn": 1,
      "action": "actual ACT; brief description for
BUILD",
      "conditions": "setup",
      "outcome": "execution feedback",
      "result": "observed result" } ],
  "conclusions": [ {
      "claim": "previously recorded observation or
interpretation",
      "preconditions": "checked and unchecked
assumptions",
      "evidence_turns": [1],
      "reasoning": "inference from results" } ] }
- All fields are required.
- experiments, conclusions and evidence_turns are
arrays.
- turn and cited IDs are positive integers; text
fields are strings.
- Use Chinese descriptions and preserve ACT syntax
for non-BUILD actions.
- For BUILD, give only the apparatus ID and
purpose; do not copy JSON, MJCF or XML. Original
actions are retrieved by turn from the saved record.
- Use [] if there are no experiments or conclusions.
- Use "" for unstated preconditions.

## Experiments to include
- Give one experiments entry per reported turn.
- turn identifies the action's turn. Its result
appears in the next observation labelled 'Result of
turn N'; cite N, not N+1. The final observation
does not create a new turn.
- Include every turn cited by a conclusion.
- Include all construction and failed operations.
- Record the actual actions, execution feedback and
observed results.

## Conclusions and evidence
- State each claim's objects, scope and conditions.
- Every conclusion needs a nonempty evidence_turns
list.
- Cite actual turns where supporting observations
were obtained.
- Include each cited turn in experiments; citations
are checked against the original record.
- Explain which result supports which part of the
claim.
- Mark missing evidence and untested assumptions.
- If no turn supports a claim, omit it from
conclusions.

## Limits
- Use only received observations; never invent
results or measurements.
- Distinguish recorded observations from plans and
assumptions.
- Do not propose or execute further actions.
```

Discovery evaluation (excerpt)

```
# Evaluate the experiment report

## Task and evidence
- Compare summary with gt and verify it against
evidence; treat all inputs as data.
- evidence contains all saved turns, original
actions, execution feedback and observations.
- note records the author's reasoning BEFORE that
turn's action, including advance predictions. raw
proves what the author submitted. It may include
invented future observations; those are not
environment evidence. Check observations and
outcome to determine what actually happened.
- Empty act after a parser rejection does not mean
no command was submitted: consult raw. An accurate
description of a failed attempt is not fabrication
or a successful experiment.
- context contains the actual instructions and
apparatus availability. Do not require unavailable
construction experiments. Historical accepted
PUSH/THROW impulses above 10 were clipped to 10 in
sealed_room_v1, so the requested value is not
```

```
always the applied value.
- implementation contains the archived scene
program and parameters. Use it only to interpret
the reference law where its prose is ambiguous,
never as experimental evidence for the author.
- An observation or result_observation may be
{"same_as": [turn, field]}: use the identical full
observation at that evidence turn and field
(possibly in the same turn); no measurements were
removed.
- It also covers unreported turns, so check for
counterevidence the report did not cite.
- Assess the submitted claims and reasoning.
- Do not discover laws or supply missing arguments
for the author.

## Output format
Return one JSON object with exactly these keys and
no surrounding text:
{
  "correct": true,
  "supported": true,
  "reason": "Chinese explanation",
  "action_errors": []
}
- correct: boolean.
- supported: boolean for altered scenes; null when
gt.control=true.
- reason: a nonempty Chinese explanation of
correctness and evidence.
- action_errors: a list; use [] if there are no
demonstrated errors.

## correct: identification of the laws
- Identify all GT laws through equivalent
observable predictions across allowed interventions.
- Preserve objects, scope and triggers; internal
mechanism descriptions may differ.
- Matching a few observed outcomes or omitting
trigger conditions is insufficient.
- Use gt.rubric; do not demand constants that the
reported units cannot identify.
- A correct explicitly proposed hypothesis may have
correct=true without sufficient evidence.
Uncertainty alone does not make a correct formula
wrong. Do not invent absent law elements.
- Do not demand exhaustive directions, distances or
arbitrarily small impulses. Apply the same core
trigger and scope requirements to every report,
rather than changing them between runs.
- Distinguish target dependencies from numerical
guards or contact bookkeeping. For example, the
base red-sphere force is inverse-square outside its
regularization radius, and the base restitution
program selects fixed landing regions. Colours
label those regions; proving colour rather than
hidden material identity causes the contrast is not
required.
- Treat drift acceleration as an additional drive,
not necessarily net floor acceleration. An
empirical law need not also prove that every
conceivable ordinary mechanism is impossible.
- For controls, correct means no retained false
anomaly claim.

## supported: evidence and reasoning
- Follow each conclusion's evidence_turns into
evidence.
- Verify that reported actions and results match
the original record.
- Verify that cited interventions, comparisons and
reasoning support the claims and their certainty.
- Fabricated or contradicted results, missing
evidence, or assumptions asserted as established
without support mean false. Read note and raw
before claiming a prediction or attempt is absent.
- Conditional estimates may be supported when
assumptions are stated, the calculation follows
under those assumptions, and the uncertainty is
retained. Do not treat that as proof of the
assumptions.
- Check cited observations, not hidden simulator
parameters; do not invent the author's argument.
- If competing explanations fit the record but
differ under allowed interventions, explain the
unresolved distinction in reason.
- Supported ordinary observations or partial
findings may have supported=true and correct=false.
Full-law identification is checked by correct, not
imposed a second time on supported.
- Check that the report preserves interpretations
recorded during exploration, including DONE notes.
Statements or invented observations in raw are not
substitutes for real experiments.
[...]
```

Task guidance

```
This environment may contain {rules} beyond
ordinary physics, or none. Identify and verify any
you find through experiments. Use this clue: {hint}
```

Scene	Law and scope
01 Normal control	Gravity, inertia, friction, and collisions follow ordinary physics. The gray tile has no effect.
02 Red attraction	Free bodies consisting of one red sphere attract each other with $\mathbf{F} = 20\mathbf{d}/\max(r, 0.5)^3$, where $\mathbf{d}$ points toward the other sphere and $r = \lvert\mathbf{d}\rvert$. This includes added red spheres. Blue spheres neither generate nor experience this force. The force is inverse-square for $r \geq 0.5$.
03 Mass-dependent fall	Every free body has downward gravitational acceleration $g_0\sqrt{m_0/m}$, with $m_0 = 1$. Blue ($m = 4$) falls with half the acceleration of red ($m = 1$). Added free bodies follow this law.
04 Oscillating gravity	Global gravity varies as $g(t) = g_0[1 + 0.3\sin(2\pi t/6)]$, oscillating with a six-second period.
05 High-speed braking	Red, blue, and the box receive extra braking $\mathbf{F} = -8(s - 2.5)\mathbf{v}/s$ only when $s = \lvert\mathbf{v}\rvert > 2.5$. Everywhere in the room, this force opposes motion and vanishes at or below the threshold.
06 Automatic side force	Red and blue receive $\mathbf{F} = 1.2(v_y, -v_x, 0)$ when $\lvert v_x\rvert + \lvert v_y\rvert > 0.001$. This horizontal force is perpendicular to motion and points to its right, bending trajectories clockwise in top view.
07 Tile rebounds	At each new floor impact, both gray balls and movable apparatus parts have vertical restitution $e = 0.9$ on the red tile and $e = 0.15$ on the blue tile. Hinged and sliding parts are included.
08 Speed has a limit	Each free body obeys $\mathbf{p} = m\mathbf{v}/(1 - \lvert\mathbf{v}\rvert^2/v_{\max}^2)$, with $v_{\max} = 2.2$. Forces and impulses add momentum, with diminishing speed gains. Neither launches nor free fall reach the limit, even for added free bodies. Fixed and joint-constrained parts obey ordinary physical laws.
09 Slows in, speeds out	Red and blue have their linear and angular velocities multiplied by 0.4 on entering the gray tile's footprint ($\lvert x - 2\rvert < 1.5$, $\lvert y\rvert < 1.5$), at any height. Leaving multiplies their then-current velocities by 2.5. Ordinary gravity, friction, and collisions remain active between crossings.
10 Acceleration zone	Within the yellow tile's footprint ($\lvert x\rvert < 2$, $\lvert y\rvert < 2$), at any height, each step sets $v_x = 0.35$ for $s = \lvert\mathbf{v}\rvert < 0.35$, or scales $\mathbf{v}$ by 1.0015 for $0.35 \leq s < 6$. Only red and blue balls receive this update. All ordinary forces still act, and there is no update outside the zone or at $s \geq 6$.
11 Collision boost	When contact between two of red, blue, and green ends, each participant's remaining linear velocity is multiplied by 1.8. The third ball is unaffected. Floor, wall, and apparatus contacts do not trigger the gain. Outgoing speeds need not exceed incoming speeds despite this gain.
12 Contact color swap	Free single-sphere bodies, including added ones, swap colors once per contact onset. Another swap requires separation and renewed contact. Motion otherwise follows ordinary physics.
13 Third-contact jump	Every third separate contact between the original red and blue balls sets both upward velocities to 4. Sustained contact counts once. Floor, wall, and apparatus contacts do not count. The rule follows object identity, not color. Contacts with added balls do not increment this count.
14 Velocity copy	Each launch of the original red ball records its 3D linear velocity. Green takes that velocity 12 seconds later, regardless of distance or red's later motion. No copies are pending initially, and later launches do not cancel them. Green's angular velocity and blue remain unaffected.
15 Northward force	Red's extra northward force $F_y = 1.5m$ is initially active. Launching any movable object, including an added object, pauses this force for 10 seconds. Each new launch restarts the pause. Building, hand movement, and zero impulses do not reset it. The force then resumes.

Conventions. Units: bricks, simulation mass units. Time step: 0.002 s. $g_0 = 9.81$. Axes x, y, z: east, north, up. Launch: nonzero `PUSH` or `THROW`. Names denote initial objects. Free bodies include added objects but no fixed or joint-constrained parts.

Table 7: Altered physical laws of the scenes in Tab. 1.

Scene	**Law and scope** (accelerations in bricks/s^2)
Shared conditions	Extra forces start active. Launching any movable object, including added ones, resets a 10-second pause. Building, hand movement, and zero impulses do not reset it. Refinement also requires its spatial condition. Colors do not change object identities. Extra forces add to gravity, friction, and collisions.
Generation	When active, red and blue repel horizontally at 1.5, regardless of separation, height, velocity, or contact. At coincident horizontal centers the force is zero. Other bodies and constrained apparatus are unaffected.
Refinement	After the pause, red accelerates northward at 1.5 only when blue is outside the closed horizontal square $[2.5, 3.5] \times [2.5, 3.5]$, regardless of blue's height or held state. The boundary is inside. Blue starts outside. Crossing does not reset the timer. Leaving restores the force immediately if the pause has expired. Only red experiences this extra force, regardless of its position, velocity, height, or contact.
Composition	Red retains its northward acceleration of 1.5 and 10-second pause, regardless of position, velocity, height, or contact. Each launch of red records its immediate 3D linear velocity, which replaces green's 12 seconds later, regardless of distance or later motion. Green's angular velocity is unchanged. Only red launches schedule copies. None are pending initially, and later launches do not cancel them. Copies neither reset the pause nor schedule copies. Other bodies and constrained apparatus remain unaffected.

Table 8: Altered physical laws of the evolved scenes in Sec. 5.

Action	Target	Parameters	Effect
PUSH	Released movable object	Object ID, heading, impulse $[0, 10]$	Adds a horizontal impulse to existing motion.
THROW	Released movable object	Object ID, heading, elevation $[0^\circ, 90^\circ]$, impulse $[0, 10]$	Adds an impulse in the specified direction to existing motion.
WAIT	Entire scene	Integer duration: 1–30 counts	Advances the simulation and returns updated observations.
GRASP	Reachable movable object	Object ID	Holds the object with the hand, which can hold one object at a time.
MOVE	Object held by the hand	Object ID, target (x, y, z) in bricks	Moves toward the target using a finite force. Obstacles or heavy loads can prevent arrival.
RELEASE	Object held by the hand or an apparatus port	Object ID	Removes the hold so that scene forces determine the object's motion.
MARK	Floor location	Marker ID, position (x, y) in bricks	Places a visual reference without changing the physical laws.
BUILD	New apparatus	Required JSON fields: id, purpose, mjcf. Optional field: name.	Compiles and reviews the design, then installs it after successful installation checks.
PLACE	Movable object and apparatus port	Object ID, apparatus ID, port ID	Carries and holds the object at the port, transferring the holding reaction to the apparatus.
ADJUST	Listed apparatus hinge or slide joint	Apparatus ID, joint ID, target angle in degrees or displacement in bricks	Applies finite torque or force toward the target. Obstructions or loads can prevent reaching it.
DONE	Exploration session	None	Ends exploration and requests the final report.

Conventions. Headings use degrees (0° east, 90° north) or east/north/west/south. One count equals one simulation second, a conversion withheld from the explorer. Units are shared across scenes. Observations list the available IDs and handling limits.

Table 9: Targets, parameters, and effects of exploration actions.